The empirical recurrence for OEIS A232926, proved by transfer matrix

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Abstract

OEIS A232926 counts arrays over $\{0,\ldots,2\}$ in which no two entries that are neighbours in a named set of directions may sum to 2, together with clauses that pick one array out of each class of arrays differing by a renaming. The entry carries an empirical recurrence of order 46 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: the clash condition is local to two consecutive rows and the renaming clauses are carried by one bounded piece of extra state, so the count is a walk count on $S = 50$ states.

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1 The conjecture

OEIS A232926 is "Number of 7Xn 0..2 arrays with no element x(i,j) adjacent to value 2-x(i,j) horizontally, diagonally or antidiagonally.". It has offset 1 and begins

$$2187, 4374, 106494, 1502736, 28178262, 459797904, 8119777890, 137439977160, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 15*a(n-1) +211*a(n-2) -2863*a(n-3) -13429*a(n-4) +203167*a(n-5) +281913*a(n-6) -7273959*a(n-7) +2536559*a(n-8) +145820567*a(n-9) -211890258*a(n-10) -1707543577*a(n-11) +3980736037*a(n-12) +11821765895*a(n-13) -39503723461*a(n-14) -45891971184*a(n-15) +241882321639*a(n-16) +65778053431*a(n-17) -978590162431*a(n-18) +246879014309*a(n-19) +2718479809492*a(n-20) -1652020665317*a(n-21) -5304510492989*a(n-22) +4695056741311*a(n-23) +7353393608676*a(n-24) -8375817160636*a(n-25) -7242062931643*a(n-26) +10268995022275*a(n-27) +4987531095051*a(n-28) -8951397463580*a(n-29) -2288267915190*a(n-30) +5615076543224*a(n-31) +596139936244*a(n-32) -2532457629816*a(n-33) -11619231656*a(n-34) +811694638348*a(n-35) -54333866496*a(n-36) -180592320980*a(n-37) +21480028944*a(n-38) +26801997448*a(n-39) -4229090640*a(n-40) -2481395056*a(n-41) +464483232*a(n-42) +126694304*a(n-43) -26501248*a(n-44) -2659968*a(n-45) +599040*a(n-46) for n>47

As of the "Last modified" line on the live entry (Jul 23 2025 07:44:07, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array is 7 cells across with entries $x(i,j) \in \{0,1,2\}$. Call two values u, v CLASHING when $u + v = 2$. Two cells are neighbours when one is obtained from the other by one of the offsets

$$(1,-1), (1,0), (1,1),$$

that is, in the antidiagonal (ne–sw), vertical and diagonal (nw–se) directions, and the condition is that no two neighbouring cells carry clashing values. The entry writes the array with n as the number of COLUMNS. The walk is run down the transpose, so the offsets above are already the transposed ones: transposing exchanges the horizontal and vertical directions and fixes each diagonal one. The clash relation is symmetric, so an offset and its negative name the same pairs and only one of each is listed.

3 The walk

Every offset above moves the row index by at most one, so a clash is a condition on two consecutive array rows and on nothing else.

So a single array row is a state. A step appends a row, which settles every clash between it and the row above and every clash inside it. Two states from which, for every remaining length, the same number of arrays can be completed contribute identically to the count, so they are identified; after that identification there are $S = 50$ of them.

With ι the indicator of the states that may begin an array and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^\top M^j \tau.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^{46} - \sum_i c_i t^{46-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+46} - \sum_i c_i M^{j+46-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 15\,a(n-1) + 211\,a(n-2) - 2863\,a(n-3) - 13429\,a(n-4) + 203167\,a(n-5) + 281913\,a(n-6) - 7273959\,a(n-7$$

for every $n > 47$.

Proof. The vector $w = q(M)\tau$ was formed by 46 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 47 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order, in the orientation the entry's own name writes, and test each clause the moment it can be tested. No transfer matrix, no transposition and no state are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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